

Yuan (Paul) Zhuang

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Research focus: foundation models, efficiency, adaptive computation, reinforcement learning, and embodied AI/robotics

Education

University of Connecticut, Ph.D. in Computer Science and Engineering, Storrs, CT	Expected Dec 2026
University of Pennsylvania, M.S. in Data Science, Philadelphia, PA (GPA: 3.87/4.00)	May 2024
Zhejiang University, B.Eng. in Ocean Engineering and Technology, Hangzhou, China (GPA: 3.65/4.00)	Jun 2022

Industry Research Experience

Microsoft Research, Research Intern, Redmond, WA May–Aug 2026

- Developed AdaWrite, a chunk-level early-exit method for frozen 1B test-time-training language models that skips deeper layers and fast-weight updates while adding less than 0.2% trainable parameters.
- Implemented AdaWrite for LaCT in PyTorch/FSDP and TTT-E2E in JAX, with multi-GPU training and evaluation on long-context tasks with sequences up to 128K tokens.
- Improved LaCT's mean S-NIAH-1 retrieval accuracy from 73% to 81% while reducing analytical prefill FLOPs by up to $2.21\times$; achieved up to a $1.38\times$ reduction on TTT-E2E.

JD.com, Research Intern, Embodied Technology Group, Beijing, China May–Aug 2025

- Experimented with $\pi_{0.5}$, OpenVLA, and NVIDIA GR00T on public benchmarks within a logistics research project focused on parcel sorting and palletizing; designed methods for multitask generalization and robustness across tasks and environments.

Selected Research & Publications

LD-MoLE: Learnable Dynamic Routing for Mixture of LoRA Experts, First author ICLR 2026

[\[Paper\]](#) [\[Code\]](#)

- Designed and implemented a Sparsegen-based fully differentiable router that learns how many experts to activate for each token and layer, with a closed-form objective for controlling sparsity.
- Achieved the best average score among Top-K and ReLU-routing baselines across nine benchmarks on three 1.7B–8B large language models from the Qwen3 and Llama 3 families.

Low-Rank Adaptation for Critic Learning in Off-Policy Reinforcement Learning, First author Preprint, 2026

[\[Paper\]](#) [\[Code\]](#)

- Introduced LoRA as structural regularization for off-policy critic learning from scratch, freezing randomly initialized residual-layer weights and training low-rank updates.
- Evaluated SAC and FastTD3 on seven DMC-Hard and six IsaacLab locomotion and manipulation tasks, with critics up to 144M parameters on DMC-Hard and 4,096 parallel simulation environments in IsaacLab.
- Improved mean return across seven DMC-Hard tasks by 20.8% over the dense baseline in the large-scale SAC+BRC setting, with 92.6% fewer trainable critic parameters.

YOLO-MARL: You Only LLM Once for Multi-Agent Reinforcement Learning, First author IROS 2025

[\[Paper\]](#) [\[Code\]](#)

- Developed an offline LLM pipeline that generates coordination strategies and Python planning functions for reward shaping; policy training and execution require no further LLM calls.
- Evaluated QMIX, MADDPG, and MAPPO on simulated task-allocation and navigation benchmarks relevant to multi-robot coordination; improved final mean return by up to 105% and achieved up to $2\times$ faster convergence in training steps.

Artificial Foveated Perception for Mitigating Shortcut Learning in Robotic Foundation Models, CoRL 2026

Second author [\[Paper\]](#) [\[Code\]](#) [\[Project\]](#)

- Developed AFP's language-conditioned mask predictor, combining CLIP-conditioned region queries with multiscale visual features to guide policy attention during fine-tuning, without requiring AFP at inference.
- AFP improved out-of-distribution success by up to 37 percentage points; evaluations covered four foundation models on eight simulated tasks and $\pi_{0.5}$ on five physical manipulation tasks.

GNN-Based Swarm Modeling and Decentralized Control, Research Assistant, Penn GRASP May–Sep 2023

- Built graph neural networks to infer interaction structure and leader roles in robot swarms, then used the learned graphs to implement decentralized controllers.

Image-to-Image Translation: Diversity-Fidelity Trade-off [\[Code\]](#)

- Implemented BicycleGAN and a Soft-IntroVAE variant in PyTorch; evaluated fidelity and diversity with FID and LPIPS.

Technical Skills

Programming/Systems: Python, C++, PyTorch, JAX/Flax, TorchTitan, DDP/FSDP, Triton, ROS, Linux, Docker

ML/Robotics: Long-context LLMs, MoE, PEFT/LoRA, test-time training, reinforcement learning, VLA models, IsaacLab

Honors

Pratt & Whitney Advanced Systems Engineering Fellowship (2024, 2025); UConn Predoctoral Fellowship (2026)